

Dynamic VLAN:

- o DVLAN is a term, which stands for Dynamic Virtual Local Area Network.
- o DVLAN is opposed of Static VLAN do not require to individually configure each port.
- o Dynamic VLAN allows Clients to connect to network based on their credentials.
- o Dynamic VLAN pushes Users/End Points into a specific VLAN based on their identity.
- o Client is place in appropriate VLAN irrespective of the ports to which they are connected.
- o Both MAC authentication and 802.1X authentication support dynamic VLAN assignment.
- o With dynamic VLAN assignment, port becomes a MAC VLAN member of the specified VLAN.
- o After authentication succeeds, VLAN assignment policy can be applied to control the client.

Create new or existing authorization profile to assign VLAN. Navigate under "**Policy > Policy Elements > Results > Authorization > Authorization profiles**" and click "**Add**" button to create new profile or click on already existing profile. Configure VLAN under profile:

The screenshot shows the Cisco Identity Services Engine (ISE) configuration page for an Authorization Profile. The navigation path is highlighted with red numbers 1 through 8:

- 1: Policy (in the top navigation bar)
- 2: Policy Elements (in the sub-navigation bar)
- 3: Results (in the sub-navigation bar)
- 4: Authorization (in the left sidebar)
- 5: Authorization Profiles (in the left sidebar)
- 6: * Name (input field containing "MAB-Profile")
- 7: VLAN (checkbox in the Common Tasks section)
- 8: ID/Name (input field containing "100")

The main configuration area for the "MAB-Profile" is shown. It includes fields for Description ("This is MAB authorization Profile"), Access Type (ACCESS_ACCEPT), Network Device Profile (Cisco), Service Template, and Track Movement. The Common Tasks section includes checkboxes for DACL Name (MAB-DACL), ACL (Filter-ID), and VLAN. The Tag ID is set to 1, and the ID/Name is set to 100.

Now assign this profile to MAB Authorization Rule. Go to **Policy -> Authorization**.

Already done above in DACL Lab.

SW# **show authentication sessions interface ethernet 0/0**

SW# **show ip interface ethernet0/0**

SW# **show vlan br**

VLAN	Name	Status	Ports
1	default	active	Et0/1, Et0/2, Et0/3, Et1/0 Et1/1, Et1/2, Et1/3
100	admin	active	Et0/0

SW# **show authentication sessions interface ethernet 0/0 details**

```
Interface: Ethernet0/0
MAC Address: 000c.2955.68e0
IPv6 Address: Unknown
IPv4 Address: Unknown
User-Name: 00-0C-29-55-68-E0
Status: Authorized
Domain: DATA
Oper host mode: multi-domain
Oper control dir: both
Session timeout: N/A
Restart timeout: N/A
Periodic Acct timeout: 300s (local), Remaining: 93s
Session Uptime: 510s
Common Session ID: C0A8509600000000E0004CB2A
Acct Session ID: 0x00000002
Handle: 0xA3000002
Current Policy: POLICY_Et0/0

Local Policies:
Service Template: DEFAULT_LINKSEC_POLICY_SHOULD_SECURE (priority 150)
Security Policy: Should Secure
Security Status: Link Unsecure
```

Server Policies:
Vlan Group: vlan: 100